

MLDawn "Brain in the Dark Design Principles for Neuromimetic Inference under Free Energy Principle"

Source: [YouTube](#) Playlist: 4th Applied Active Inference Symposium (2024)

Duration: 1:03:55 | Uploaded: Unknown Transcript method: auto_caption

hello welcome back it's the fourth applied active inference Symposium day two part two on November 14th 2024 we're kicking off this session in the darkness of the preon on the west coast appropriately with Maron Hussein Zade bagani ml Dawn brain in-the-dark design principles for the neuromatic inference under the free energy principle so thank you Maron for joining looking forward to your presentation thank you very much it's a pleasure to be here um so today I'm going to be talking about uh the design principles for neuromatic inference under the free energy principle namely the methods that are inspired by the brain um in order to model human perception um just um want to preface this by saying that this is a collaboration with my colleague Simon orbas from Manu University and Professor friston from UCL and this work has recently been accepted at the neuro AI workshop at neuros 2024 so let's just get started the content is going to be as follows we're going to be talking about first the motivation behind this work why we got started started with this idea anyway to to introduce the design principles particularly for these models and then I'm going to do a quick refresher on the math side just so we all get comfortable with the you know the upcoming derivations that I'm going to be uh expanding on which covers the math behind free the free energy principle particularly a variational free energy then I'll do a brief discussion on what free energy principle is I'm presume Knows by now because we we' had so many talks uh since uh the beginning of the Symposium but I'm going to do a quick run at that then we'll be talking about the beian brain hypothesis that furnishes the foundation for the idea of modeling human perception as a kind of inference namely basan inference next we dive into dejos between the generative model and a generative process what do we mean by each one of them knowing the distinction is crucial for understanding the free energy principle because Fe lies on this um notion that there is a boundary that separates an entity from uh the environment and separating uh basically my internal state from the external State beyond my marov blanket next we'll talk about

variational free energy and how it's driven how it's derived basically and then we jump into the design principles in terms of the experiments not the things that you need to consider the things that you need to Define and how you should go about doing that in order to model human perception as a kind of online inference problem we'll go through some final thoughts some lessons learned and then I'll introduce some of the resources for those of you who are interested in uh you know pursuing free energy principle more seriously and delving into the theory and the Practical side of it and then we're going to wrap it up with uh a demonstration um using the GitHub repository that that is publicly available by um by the way so the motivation is this so you said deep learning has revolutionized artificial intelligence um because it has automated the process of feature extraction from raw data however it actually faces a lot of problems for for example lack of outof distribution generalizability catastrophic forgetting and poor interpretability it is known as a black box basically and the catastrophic forgetting is basically when the new knowledge overwrites the previous knowledge the weights are getting overwritten so while the network learns the concept of a cat it might momentarily forget the concept of a dog because the paths are shared between different Concepts that are to be learned you see in contrast to deep learning biological neural networks in your brain and a my brain they do not suffer from these issues and this inspires the AI researchers to actually explore this interesting and intriguing area called The neuromatic Deep learning that is all about replicating the brain mechanisms namely the neuronal message passing and belief updating within the AI models purely sprayed by our best knowledge about how the brain works um a foundational theory for this approach is the free energy principle um Fe which unfortunately despite its potential is often considered too complex to understand and to implement an AI simply because one would need a myriad of disciplines um under their belt knowledge about disciplines under their belt to be able to understand the free and principle namely you need to know things for example about dynamical systems random processes you need to know about basian mechanics some Neuroscience some maybe computational neuroscience and so on and so forth so as a as a result it's not it's an intimidating topic let's say for anybody who wants to just get started with it so we seek to demystify FP and basically provide a comprehensive framework for Designing neuromatic models within well where we are actually trying to mimic humanlike perception capabilities um so here we present a road map for implementing these models and by the way a pytorch code repository for applying FP in a predictive coding network is going to be introduced here the link is here that that's a code it's publicly available and also as I said the paper will soon be

available um by the neuro AI Workshop um and so hopefully you guys will have a chance to take a look at that as well now let's get started with the ref refresher of the math so first foremost the idea of a PDF so it it's an acronym for a probability density function which is a function that describes the likelihood of a continuous random variable taking on a particular value it has two conditions it has to be positive so for a given parameter θ_0 for example the mean of a gaussian let's say for any input X it's always positive and second we have this normalizer uh sort of condition that if you integrate a PDF across the support space of of the input name the all possible values of X you will inevitably must receive get to the actual value of one what this means is the probability of X the input getting any value in a in any range within its support is always one which means that X will always fall in in a range in that in its own accepted range of values so that's 100% so you need these two if you want to have a PDF next is the idea of an expectation I just want to remind you that the reason I'm going through this math is because later on when I introduced the steps of deriving um you know variational free energy these concepts are absolutely crucial so hopefully um I'm not going too fast but um yeah just bear that in mind that these are very important and for those of you who are you know math Savvy and these are two Basics my apologies now the idea of expectation is actually the weighted average of all the possible values that the variable consume weighted by their probability so you have the in the first line um the expected value of x over the probability distribution the PDF P_X so as you notice we're integrating all possible values of X multiplied by or weighted by their corresponding probability and it's important to note that you don't you're not limited to you know calculating the um the expected value with respect to a variable but you can also do this exactly the same thing with a function so the expected value of function is again the same thing but you are summing up a weight you're basically calculating a weighted average of a function where the weights are the probability of the input to that function which is X now some key Probability components here now we have this idea of a joint probability um so a joint distribution between two variable X and Y you can divide it up between like like some conditional term which is the probability of x given y multiplied by the probability of Y in the first place or you can go the other way around the probability of Y given x times the probability of X you notice that if X and Y are independent this conditional term reduces to probability of X or this conditional term would reduce the probability of Y so either way on both sides you're going to have the probability of x given y um if X and Y are indeed independent next if want to move from a joint distribution to a marginalized distribution where you're actually marginalizing one of the variables out this is

how you can do it so in this case if you integrate P of XY across all possible possible values of X in other words like you literally break this down and you know basically write it down in in this you know um constituent terms if you do this integration you're inev what basically what you're doing is you're getting rid of X and then you end up with a term with a probability distribution that is only a function of um Y and that's what you call a marginalized probability distribution or PDF again this is key where will this come in handy it's going to come in handy when when I'm going to be talking about the basium brain hypothesis and base Rule and so on and so forth so keep these two equations in mind please now some properties of logarithms so this is basically just the definition of a log so log of a with base b equals c the definition says that it just means a equals B to the power C right so for example the log of 8 base 2 equals 3 because 2^3 is 8 so the log of multiplication is some of the logs of individual terms the log of a division is the subtraction of individual log terms that log of denominator minus the log of the denominator um the log of a fraction is literally minus the log of the swapped fraction literally swap b and a together from like you know you end up with just a negative of the of the new log term um the power law the log of B to the power C so basically C drops down as a coefficient so you end up with this term $C * \log$ of B with base a log of a with base a if the value and the base are both the same you end up with value one and by extension if you have this term C drops down as a coefficient and this term equals one so you end up with $C \log$ of one with any base is always equal to zero why because a whatever it is to the power zero is one and last but not least this one is again very important $\ln$ of a is basically just a logarithm a logarithm with you know where the B is natural number e or it's called the natural log and E is basically the oiler Oiler number so you got log of a with base e it's called just the natural log so when in in Python when you say numpy do log is basically the natural log now the free energy principle it is a theoretical framework in Neuroscience it's been developed by Carl friston so basically things that persist over time especially living systems look as if they are minimizing a certain quantity it's called free energy it is as if they're uh sort of maintaining this boundary that is called a mark of blanket to persist in the face of random fluctuations in the uh in the environment um and this really connects well with the beian brain hypothesis you as you will see shortly so the brain from the perspective of beian brain hypothesis it's actually in the game of inferring the hidden causes behind its Sensations so namely if you have if you let's say y to denote the sensations and X as the hidden State and then this inference basically means that given my observation why I want to estimate the probability distribution over X right it's basically called a posterior probability which means that after having seen an

observation um what is causing in the world beyond my Markov blanket this particular sensation or observation inevitably this requires having a generative model GM of the world and this actually connects really well with Helm hold's um concept of unconscious inference and it is defined like that perception is kind of unconscious inference which means that we are in the game of you know inferring the hidden cause behind our our Sensations for example right now you're looking at the Monitor and the photons but the light is hitting the your sensory epithelia on your eyes those excitations those those hits those are your sensation and when you realize that I'm I'm watching this presentation at at at active inference Institute um and this guy is talking his name is mean this is the slide of a of a face Helm holes pictures here these are the inferences that your brain is making given that particular Sensation that is hitting the the neurons on the surface of your eyes right so it's estimating this probability distribution about what's happening beyond your markup blanket now let's talk about the idea of generative model versus generative process this is actually very very important on the right hand side you see this IDE this generative model as a box and it has some internal States let's call them hidden States this represents the neuronal activities in your brain um the excitabilities um different you know the the voltage the voltage level let's say of your neurons that are keep changing all the time they fluctuate all the time and these guys I BAS basically holding information about what's happening beyond your Markov blanket right remember what's happening beyond your Markov blanket is out of your reach you have absolutely no way of observing exactly what's happening right and here we have the generative process where xstar denotes the actual things that are happening in the world which are completely out of reach to you now if you imagine here you have the face of a person let's say steep job is standing in front of me but this is no like I'm I'm not going to see the truth what I'm going to see is some sort of a um a map some sort of a noisy transformation of the truth xstar right let's call it why this is what I'm going to be experiencing this is what's going to hit and uh induce excitation at my sensory epithelia on my eyes and then if you want to look at like um basically dig deep into the idea of perception as um unconscious inference the G the game is this your brain the generative model in your brain predicts a sensation like I'm expecting this sensation to be felt or received at my sensory epithelia on my Markov blanket and then the actual sensation why is received so the difference between the actual sensation and my prediction is called the prediction error so using that prediction error I will update my G model about the world namely I will update my belief about what's happening in the world so this is where my internal States will change to reflect a better image of the unknown World beyond my markof

blanket or the parameters of my generative model are going to change um to again do a better estimate of what's happening in the world so that I will have a better prediction about the sensations that that world is going to cause right so generative process again is not directly observable and the internal uh States X this is very important carry information about the invisible external States x beyond my mark of blanket this is where this connects to the IDE the discipline of information Theory this is a very important link by the way so the generative model what do we mean by that mathematically speaking now in a dynamical world a generative model has belief about the Dynamics of the world what it means is how fast is the world changing for example or how is it changing now if the world is at a given State X right what does its Dynamics look like for example it's velocity or acceleration or even higher temporal derivatives are like jerk basically all sorts of rates of changes the rates of change about the world that's what what I mean by Dynamics so in a generative model your generative model will have an understanding has to have a belief about how the world is changing Beyond a smart C blanket so in other words if I'm in the game of estimating the Dynamics of the world it means I'm chasing a moving Target what it means is at any given time when I want to infer what's happening in the world by the time I've made that inference the world has already moved so that basically makes inference uh a tricky task especially if I'm in a in a you know highly dynamical World it makes it more challenging the second thing that in a Gena of model we should have is the Gena of model has a belief about how the world causes its Sensations namely if the world is at any given state that by my estimation beyond my Markov blanket what is the most likely sensation why that I will be experiencing right so these two are the key components of a generative model so like mathematically this is how you show it $\dot{x}$ is literally the velocity of uh the world how fast it's changing mathematically it's just DX over DT and then you have some function of the states X and causes new and Theta the parameters of my generative model plus some random fluctuation ΩX so that equation is called a state Dynamics equation but what it basically is telling you is that the Dynamics of the world is driven by X by the position of the world basically and these are all my belief about how the world is changing DX over DT the velocity of the world and technically speaking f is called the flow of the state Dynamics it's the deterministic part of the state Dynamics ΩX random fluctuation is what encodes your un certainty about your belief about how the world is changing yeah the second equation uh basically connects to the second concept that I introduced here it's basically telling you that my belief about my sensation about like what's the most likely um Sensation that I'm going to be receiving is

encoded in this equation where my observation is equal to the G of the function of again hidden States the causes new and the parameters some parameters plus some random fluctuation Ωy it's called the observation model again Ωy encodes your uncertainty about your belief about the most likely Sensation that you will be experiencing a sensation that is caused by the world you notice that we have this these again your hidden state X and then uh the cause is new now if you're wondering so you know what the thetas are these are just the parameters of the functional form of FNG G but if you're wondering what causes are this is something that I'm not going to be focusing in this um particular talk but just imagine that if you have a hierarchical generative model these causes are just a subset of the Hidden States these are like the hidden states that do not have Dynamics but they're the tools that different levels in the hierarchy use to be able to talk to each other so interlayer communication is done using these news or the causes these are the ones that basically Drive the Dynamics you notice that that's why we have $x = f(x)$ and new they they drive they drive the Dynamics without them the layers cannot talk to each other basically now if you want to like turn this into a belief-based system where you have probabilities in play you just need to assume a gaussian distribution over your random fluctuation namely with mean mean zero and some Precision uh over your random variable X and your observation y just my you know Open brackets Precision is just inverse uncertainty um and close brackets that's just a notion in statistics Now using this if you literally replace um you know this um notion of a Gaussian distribution into your original equation you end up with this probabilistic sort of um uh you know notion that the probability of X do given all of these three elements is distributed according to our normal distribution where again the Precision is the same as the Precision of your random fluctuation but you notice that because we're summing up f with this guy over here the the mean of that Gaussian distribution becomes this F term same story for the your belief about your why your your observations and that is why I'm saying that $f(x)$ and new and Theta is the expected your expected um Dynamics about X do because it's literally the expected value of the gaussian distribution same story here G of X new and Theta is your expectation about your observation your sensation again it is the sort of expected value of this gaussian distribution the value that is most likely to occur so that's generative model two things we know the Dynamics of the world we have beliefs let's say we don't know it we have beliefs about the Dynamics of the world and we have belief about uh basically the most likely Sensation that we will be experiencing if such and such was the State of Affairs of the world let's say x was the State of Affairs in the world now we talked about inference let's talk about

Base rule really quickly now here this is base Rule and the components of Base rule are as follows we have the likelihood term which is the probability of my observation given uh the state of the world we have a prior belief which means that before observing why this is my belief about what the world looks like we have the model evidence which is probability of data and these two together is called basically the the generative model again like this is in line with what we discussed here this is basically your prior belief about the Dynamics and this is literally the mapping between X and Y which is your likelihood term so I'm not saying anything new here these two together are your generative model and this is your model evidence this is a devil right this is intractable you notice that in a real life scenario we will not be able to calculate this and this term on the left hand side is called posterior probability it's your belief about the distribution over hidden state of the world X and when you map your generative model using base rule to uh this posterior distribution we call this model inversion right so if whenever you hear model inversion in the context of FP that's what we mean by that so this evidence again going back to our idea of marginalizing a variable out if you again just remind yourself that this is how you marginalize the hidden State X out you just end up with probability of let's say sensation why you notice that this integration could get really nasty really fast which means that if you have you could have basically any number of hidden States X and for each X you would have a range of possible values so this thing could turn into a potentially like potentially in infinite number of nested integrals basically it can become intractable really really quickly so what this means is we will not be able to calculate this posterior probability um exactly so what should we do the answer is variational inference so because U you see probability of data your model evidence is intractable but at the same time ironically it's the true measure of how good your generative model is um which means that we call this also marginal likelihood right so if it is high it means whatever sensation you're experiencing right now under your model of the world under your generative model of the world the probability of that particular sensation is very high it means under your model of the world everything that you're sensing you already expected to sense it so it means you have such a strong and uh you know and precise belief and precise yeah belief about the Dynamics of the world and how the sensations are generated from those uh those dynamics that whatever you're receiving is not surprising that that's why model evidence is such a strong metric for evaluating how good a generative model is but unfortunately it's intractable so can we find a bound to this namely a lower bound and the answer is yes I'm going to introduce the idea of a evidence lower bound and the next slide so by doing this I'm

turning the problem on its head so I'm turning a converting let's say a an intractable exact calculation of the posterior and I'm replacing that with a tractable approximation um well basically a tractable approximate estimation of the posterior and that's called variational inference and I'll tell you what why we call this variational by the way because the objective function that we're going to be minimizing to do that is VAR free energy is actually a function of a function it's a functional so where wherever you're dealing with functionals and optimizing functionals U that's where the calculus of variation comes into play that's what what it means by variational inference now on the right hand side I'm just going to put some reminders the expected value and the this particular particular law of logarithms and in the middle this is called Jensen inequality just keep keep them keep an eye on this one it just says that the expected value of a log of something is always a lower bound to the log of the expected value of that thing right so this is going to come in handy so let's study the this U amazing concept of evidence lower bound remember P of Y is intractable we want to find the bound to it so this is log of probability of data um this is how we mark marginalized X out and you can multiply and divide by this by by by something right whatever it is so here QX is we call this surrogate the surrogate posterior over the hidden State this is going to be a representative or your approximation to the true unknown posterior belief right so let's call it QX and it's by the way it's really a key factor in variational inference and Fe so keep that in mind this Q um um variable this Q function so moving on you you do a little bit of play here so this this sort of Cu moves to the denominator and you notice that this is the definition of expectation with respect to this particular distribution Q right just getting the definition from uh from here and according to Jensen's inequality um this term has a lower B and this is the lower bound to it so the expected of of this term the expected value of this term over this probability distribution you have the joint here and you have the surrogate distribution here and you see so this right hand side again remind yourself this is your log model evidence and we found the lower bound to it using Jensen's inequality right so the lower bound is found this guy is your evidence lower bound um and that that's a key finding that by maximizing this evidence lower band you're actually maximizing uh your your log model evidence and you're getting closer and closer to the true posterior now elbow and vfe just want to make sure that we're clear on this we have elbow here um but VF or variational inference is just negative of your elbow so if you multiply NE minus one by both sides of this you end up with this term always bigger than minus log probability of data this is called surprisal and this is called variational free energy that's why we say VF is always an upper bound to surprise so

maximizing elbow or minimizing VF they both mean the same thing and um this is basically called variational free energy and that's why we call this a functional because this thing is a function of a of another function and this called surprisal or surprise and just shows you in a plot that here you've got variational free energy that's always an upper bound to the true unknown surprisal or by extension you got your elbow here that's always a lower bound to the true um log model evidence so VF has multiple incarnations it comes in different form by playing with the math so if this is your this is your vfe just expand you can just expand the joint and the denominator and then literally use the log rule of the you know uh the division turned into the subtraction and then you can distribute the expectation across the two different terms and this um this basically results into like just basically swapping these two terms uh by each other you end up with this equation right and these terms have names so this one actually this expected value between your belief about the the hidden State and the true unknown posterior this expectation is called a a KL Divergence it measures how different your belief about X is from the true unknown posterior that's why it's called an approximate error approximation error and the second term is your surprise right um so um if I want to like show you like what this actually means visually this is really interesting so this is your surprisal right and on top of surprisal you're adding the imation error this is this sort of length over here and then this constitutes your free energy what this means is like free energy is always an upper bound on surprisal and the error the that Gap is your approximation error is how far your belief about X is from the true posterior so by tuning your generative model your variational free energy goes down and when it goes down this Gap is reducing so which means that your Q gets closer and closer to the true true posterior which means a better and better perception about the world and this doesn't end here your free energy will get closer and closer to the surprisal um basically by extension to the model evidence which is the metric that we're going to be use using for you know comparing different models with one another a second incarnation of vfe is again we're going to expand the denominator again into this joint so previously was P of x given y times probity of y now we go $\text{pro } y \text{ given } x \text{ times } P \text{ of } X$ and again Distributing um the log and then we distribute the uh you know the expectation and then just you know uh swapping the terms again you have this scale Divergence over here and you have this negative negative expectation now this thing over here is called accuracy it means how accurate am I under my belief of the world the expect the expected value of observing why under my estimation of where the world is X so the higher this is it means the more accurate your belief about the world is that is that is basically causing your observations but at

the same time this term over here is called complexity it means that if you have some prior belief about the world you will not be able to just move your queue wherever you want it is it has to stay close to your prior belief over here so this is basically a reg regularizer if you will um and the third Incarnation is um basically again here this is the definition of vfe again you know the log rule um so we end up with these two terms so this guy over here is called expected sorry it's called expected internal energy and this guy is your entropy so this the expected value of log of QX we call this entropy um how do we minimize raal free energy we have to maximize our entropy and we have to maximize our internal energy now this term makes sense that we need to maximize it but what does this one mean it means that your Q will stay as smooth as possible as simple as possible this connects really well with aom's Razer if you're from machine learning background that your the best model is the one that provides the most accurate account of the data but at the same time the simplest explanation for that data this Simplicity is enforced by this entropy term it it states smooth so this is very interesting so these were just three incarnations of variational free energy and the goal of inverting the generative model is basically we the GM strives to minimize this vfe um this achieves two things I hinted this uh just a little bit previously but it means that the actual uh value of VF will approach the intractable minus log probability of data this is going to be used for model comparison and the surrogate distribution will approach the unknown posterior belief remember that was the goal of um you know perception in the first place and once you reach that basically minimize your variational free energy you can use the actual value of free energy as the metric as a proxy for the unknown model evidence to compare different generative models together right so the model with the lowest rational free energy is going to be the best model yeah I think I've explained that one now just to let you know that the true form of the posterior is unknown and you see the the the terms like the the probability distribution The Joint distribution here um it is not guaranteed that um you know they're analytically simple you know to to derive and and you know work with because of that um there is this idea that we can actually use llas approximation to break down each one of the distributions and basically approximate them using a tailor expansion um particularly up to the third term which means that I'm providing a quadratic approximation using tailor expansion for each one of the components um within variational fee energy um unfortunately I don't have the time to go over the detailed derivation but suffice it to say that the result of you know s approximating all the components is that the variational free energy simplifies the dise equation for those of you who are interested at to know the details I encourage you to look at

the primer on variational plusus paper which is an amazing paper um it goes through all the details of derivation so basically free energy um reduces to log of the joint distribution and you have the posterior um covariance and then yeah n is basically the number of parameters and the states that you have and so on and so forth now predictive coding this this is basically where it gets even more interesting it is a basically a hierarchical generative model that we're going to be using to model human perception and the goal is to invert this hierarchical model so it's a powerful framework to describe how the cortex accumulates evidence from noisy stimuli um it's an application of the free energy principle it focuses on how this literally fantastic organ can perform probabilistic inference about the most likely hidden cause that has caused the sensory signal um it assumes that the brain has a generative model of the world through which it constantly infers the hidden states of the world and it can also learn the parameters while inferring the hidden state of the world so model inversion happens by minimizing variational free energy again even though it's hierarchical but the concept is stays the same so in predictive coding we have this concept of top- down predictions where predictions go from inside out about your Sensations right and then the prediction error is calculated like what I described before and then this these errors if I want to be more precise the Precision weighted prediction errors are sent back from outside in uh or they call it some up sort of Direction so that the the estimations of the world the estimation of the posterior beliefs about the world are corrected basically that's what model inversion that's when model inversion happens when the posterior estimates get adjusted and it looks like something like this we have this hierarchical model that we have the same story of State Dynamics and observation model but here we have this happening at each level and then you notice that these news these causes are the ones that link layer I to I minus one that's the important part that without these news without these causes the layers cannot talk to each other you notice that it is a hierarchical dynamical system uh basically um now how do we design an experiment um for you know for for for this kind of perception neuromatic modeling using the free energy principle I just want to give you the sort of ingredients here you the designer you first have to Define your generative process that's X that's the true uh hidden stat I should have maybe said x ar yeah x star would have been a better notion then you will generate the observations this is some sort of some transformation of the X 's you can add noise to it you can transform it the way you want but that's going to be the observations that's going to hit the Markov blanket this these are what this uh this is basically where the sensory state observ and receive those Sensations you have to Define your generative model this this part is the tricky part right the fs

and G 's in your dynamical system that's where uh everything gets really tricky how would you define them if it's hierarchical how many layers should it have basically these are the big questions you have to define the evaluation metric that how would you define different Generations of models um with each other and then you uh you have to design the online State inference Loop now this is by the way just for this case for this presentation is just about State inference and then you have to compare different generations of models this is where Bayesian model comparison comes into play now here um you basically use a dynamical system here I'm just using a Lotka-Volterra process that you know captures the Dynamics between uh Hunter and the prey together it's a two-dimensional sort of world if you will so by solving the trajectories by solving the differential equations you get to this sort of trajectory so you have 1,000 data points here each which is two-dimensional so this is the unknown states of the world right this is the unknown dynamics of the world to generate a process again it's unknown it's beyond the mark of blanket again so this is the figure that we saw before now you can assume that you got the brain here and you've got uh the generative process x_{star} on the left hand side the observations why are basically a noisy version of the same hidden states that you notice that this this is really noisy here right so the brain has to receive that and then infer the true States here right so for a one layer predictive coding Network for these particular experiments you have just a simple sort of generative model it's just one layer and then uh I'm sort of comparing two generative models here where the flow of the state Dynamics are different on the left hand side you have something that we call just it's just a pullback attractor uh we have this 2 by 2 Matrix here and we have this sort of point of Attraction which assumes that everything collapses into this world just just to be clear it means that my generative model of the world assumes that the world Beyond The Mark of blanket is attracted to a particular Center called s_{aigh} right that's what I mean by believing something about the world whereas the second flow the second generative model is a sinusoidal sort of belief it means that I believe there is something periodic about the world beyond my Markov blanket and the G 's I just kept them as identity which means that whatever new is is going to be my prediction about my sensation right so I kept the G 's identical now the evaluation metric this is very important as the inference is happening online for every observation you have to make an inference um and then predict the sensation and then calculate the prediction error and then eventually calculate variational free energy because it's happening online you can actually literally accumulate your free energy variational free energy across the entire inference period so at the end you will end up with one scalar value it's called the time integral of of your

variational free energy they call it free action right it's just a fancy way of summing up variational free energy across the entire inference period in basically it's time integral and it is it's an upper bound to the accumulated surprise so variational free energy is an upper bound to surprise and um free action is an upper bound to the accumulated surprise because free action is in and out of itself an accumulation of variational free energy and this is you have to also then Define the online State inference Loop uh I don't want to go into too much detail here but um basically for each observation you will make predictions and you Calculate the prediction errors you calculate your free energy you calculate the gradient of free energy with respect to your estimations of the posterior that's the key part and then this is where you have to update your beliefs about the world and you notice you notice that it's a differential equation because the way you're going to update your um your belief about the world it's continuous it happens in continuous time it means between each two observations Y_i and Y_{i+1} I have to integrate over this update rule to move from my previous belief about the world μ_{i-1} to a new μ_i at the next time step and that's why the integration happens here and just so you know this equation is basically saying that the amount by which I'm going to change my belief about the world μ_i is first of all the gradient of free energy with respect to μ_{i-1} right that's typical gradient descent but the second term it serves as a momentum it means that if I'm going to update my belief about where the world is I'm going to consider the velocity of the world as well right if I'm going to estimate the velocity of the world I'm going to consider the acceleration of the world as well it's it's similar to the idea of momentum in optimization algorithms this is the bit that helps you you know catch up with the chasing Target that is the world that I talked about in the beginning and then by you integrate this differential equation to to update your belief about the world and then the next observation happens so this continues for all the observations and you're accumulating your free energy as time goes by and the result will be your variational free action so the result is this so this is the first generative model with the pullback attractor um so this is the uh inferred state right and this is the this is interesting this is the accumulated free action right um very very interesting you notice that free action um is having these regular jumps it means periodically I'm getting surprised about my Sensations remember that if you note here the this is where these nonlinearities happen it means that my simple pullback attractor fails to capture those Dynamics so it gets surprise but interestingly where things are linear and simple I'm not getting much surprise so it's almost constant here so I'm good at capturing simple Dynamics but not highly not the ones that are nonlinear whereas the sinusoidal one you notice that we are capturing the magnitudes much

better we have a steady increase in free action we don't get surprise for some Dynamics and less surprise for other Dynamics we are always equally surprised for everything which is interesting that like here we are getting surprised consistently for everything but here we are not getting much surprise for these particular Dynamics so one is good Capt good in capturing nonlinear Dynamics periodic Dynamics the other one is better in capturing simpler linear Dynamics which is interesting and notice that the reaction here is much lower than the freeaction here so basically this means that this model is much better now looking at the Gen the generative power of these uh GMS so this is the G function if you remember G of X and mu that's the prediction of your one layer predictive coding Network about the sensation that's the bit that gets compared with the true sensation and the prediction error is calculated remember that observations just like hden say are two-dimensional so this is the first dimension of why the true sensation and why had the predicted sensation you notice that they're almost close and also with the second um Dimension they it's not doing a bad job right but for the second generative model again it's doing a much better job in capturing the magnitudes of this of the system again you notice that it's not so good in generating these bits whereas this the first model is better in generating prediction more accurate predictions of the of the first St now these are the results the pullback attractor and trigonometric the sinosoidal um sort of flow you notice that the freeaction of the second model is lower it's better and M loss which basically comp uh compares the estimated X and the true X the inferred X and true X together is also slightly lower so free action shows that the second model is good and this is where beijia model comparison uh comes into play where we use base factor for comparison it basically says that we take the the model evidence for each one of the models and we divide them now because these are intractable we're using free action as a proxy and when you put the values of free action you get to this value because it's higher than one it basically means that you have to pick the second model because it has a lower free action it doesn't get as surprised as the first one and at some point of caution free action versus MSE MSE as a measure of accuracy should never and I repeat never be consulted on its own because it ignores complexity so for example what if the freeaction of model one and model two the model one had a lower freeaction but what if the MSE of the first model was actually higher than M than the second model which one would you choose again you have to choose model one because this has higher power of generalizability to the Unseen data so it has less chances of overfitting to the data we call that free action again as complexity minus accuracy so what if we remove complexity again we will overfit but that was important some final thoughts you

can definitely learn the parameters and also estimate uncertainty about the world so these are the things that are not covered in this presentation but you can do that this is where the separation of time scales um will come into play the demm algorithm does exactly that this is dynamic expectation maximization I encourage you guys to take a look at that algorithm it's you using that you can solve the triple estimation problem now can I go beyond x dot yes you can that's where you jump into the world of the generalized coordinates of motion where you're not only using X estimating X do but you're also estimating higher temporal derivatives of uh of the world you have a a system of couple differential equations rather than having just two equations which is very very interesting actually um also remember that active inference is not the same as predictive coding um particularly in the philosophy um you know um communities I've heard that people use active inference when they're talking about predictive coding remember that in predictive coding we are simply doing um making sense of the world there is no concept of planning or action in play um the ml Community unfortunately um does not work with the Dynamics of the world and AI uh um basically is going to the opposite direction instead of uh using smart data we're just going bigger and bigger data so we would like a world that chat GPT would also ask you a question and becomes curious about you to to receive information that it selects basically so where should you start with with the free energy principle these are the resources that I strongly recommend you some videos on YouTube hopefully you can later on pause this video and go through these links some papers that I'm recommending here and the last one the two books that I believe the red one the predictive minus for philosophy community and the first one goes through the math extensively it's a beautiful uh book yeah the uh and the videos I I I recommend that you guys start with the videos in the first place and I would like to just acknowledge our funding from the maryury and also my colleagues Simon bbus and professor priston and also acknowledge the folks that UCL UC UCD and monach University who have always supported uh uh our work and here's the link to the the code that everybody can just take a look and run the read me file is self-explanatory unfortunately I ran out of time and I won't be able to show a demo but yeah that's it so thank you Maron that was great and and uh we have 10 minutes if you want to show anything or I can ask some questions but but we have till five minutes after if you want to show anything or even just scanning the repo could be cool okay sure sure sure that's great thank you for that but just wanted to comment your your educational material and the YouTube channel are great and and they are they're an excellent supplement to the textbook and to the math learning I appreciate that yes um absolutely uh so yeah I also have a YouTube

channel mlon feel free to take a look I'm actually reading the active inference book on that channel so hopefully it would be of interest to you guys and this is the repository that is publicly available um the read me file is pretty self-explanatory and this is just a one predictive coding Network for the task of inference and modeling human perception as a task of inference these are the same pluss that you've seen um so basically all the functions are here right everything you need is here so the main function is the one that you're going to be dealing with and the yaml file is where you actually choose the parameters uh of your experiment just to show you in action what it looks like so if you actually clone this repository on your machine you just have to go to the yaml file and just do a quick scan of the readme file but basically uh using this is where you define your generative process for example you can say I want it to be a lotka vola process and this is how I'm going to be picking the DT and the total time uh to actually solve the trajectories of my generative process you can Define your generative model here you can Define the Dynamics to be like a pullback attractor or a trigonometric one the likelihood for example is identity here these are the dimensions of your X and Y two and one you might wonder why it's two for X is because you have x and x dot together where the world is and how fast the world is changing and this is just Why in your generative model you can increase them uh that's where you jump into the world of generalized coordinates of motion but this is beyond the scope of my talk today again this these bits are beyond the the scope of our talk today this is where uh you basically C talk about the correlation between the noise process across different levels of the generalized coordinates so the noise on the velocity how does it correlate with the noise process on the acceleration your belief about the acceleration of the world so how do they interact with one another that's where these um values come into play and then this is the kind of noise that you will use to alter the true states of the world and generate your observations for example you can use a colored noise here you define the mean and uh basically the standard deviation of noise and this is we're using the sort of Kernel here that convolves with the white noise to actually generate a more smooth noise the reason being that um the biological systems are believed to be uh working with uh you know to so all all the signals that you work with in biological and organic systems are believed to have been generated by some sort of dynamics that that's why we keep them as smooth noises infinitely differentiable noises rather than just white noise and uh if you actually run this so this is basically a pullback attractor this is the first uh sort of generative model that we worked in so you just run this main function so we have 1,000 observations noisy observations right and immediately this happens so for every 10

observations we are accumulating free action you notice that free action is ever increasing because it just accumulates free energy and on the right hand side you have VF or variational free energy um again if you look at the definitions inside uh the functions folder of how we calculate or compute variation of free energy you notice that um it's a far more simple simplified version of what you would see about energy is because everything is under llas approximation and you end up with far more simpler terms and the final result is reaction is this much and basically your final MSE loss is 0.53 and if you want to see the final output you just have to go into the results folder and these are the folders that are generated just so that you have two dimensions in x one dimension in y lot volterra is a generative process for example do generative model follows a pullback attractor scheme and then your G is just identity this is your free action and then that is your Ms Loss and if you open it all the plots that I showed you in the presentation are saved as PDF here just to give you an idea like this is your X this is your true invisible uh states of the world right and if you want to look at exat this is the inferred states of the world and these are for example the the noisy observations of the world yeah so I hope that the repository would be um of help and of interest to all of you thank you very much great what kinds of problems or settings do you feel that this could be adapted to or useful for um I would say first foremost this could be used for um basically it's all about any kind of problem that's about inference you have some noisy observations that you want to infer the hidden state but in terms of adaptability you can build on that by introducing parameter learning as well as uncertainty estimation so our goal is to develop this into a triple estimation sort of package predictive coding Network and add more layers to it so to have a hierarchical predictive coding uh capable of solving the triple estimation problem yeah that's the ultimate thing that you can actually adapt this cool um what are your uh what are you excited about for ML Dawn or for your learning or research in next year um I'm very much excited in embedding these kinds of calculations into the world of AI because as I said AI is moving in my opinion and my humble opinion is moving to the towards a wrong direction instead of bombarding models with all sorts of data that the model should be able to choose its data so I'm very excited in you know developing this framework into a hierarchical predictive coding Network and then sort of compare the performance of such a model with other traditional aib based generative models and basically show its superiority um that I believe it has in different scenarios and ultimately this model is going to be used for in my grant proposal and a project I'm working on is going to be used for modeling hallucinations as a kind of false inference about the world um and it's going to be used as a proxy to for us to

understand what happens in uh Neuropsychiatric disorders when people actually suffer from hallucinations in the brain so this predictive coding network is going to be a brain inspired AI model for us to understand what goes wrong in in the skull of a human being when they suffer from hallucinations about the world so that's what I'm really really excited about and really hoping to achieve cool it's like inspired by the brain for the brain absolutely thank you on great great presentation and good luck with the continuation of the work appreciate it thank you for having me okay till next time e